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INTRODUCTION OF THE "AUDIBLE MATHEMATICS" SOFTWARE PACKAGE

"When introducing a new concept, we do it correctly if we consider several
we use illustration."
Zoltán Dienes

Summary

Audio aids for visually impaired students[22]

Supplementing the information has a positive effect not only on the development of auditory attention, auditory memory and analytical skills, but can also contribute to the strengthening of many basic mathematical competencies.

We intend to expand the tools supporting the integrated education of blind and visually impaired children with the software we have developed. In developing a digital toolkit that can also be used for sighted lower-grade students, we tried to take into account the experiences gained among the primary target group, the latest results in psychoacoustics, and the currently applied didactic methodologies.

Keywords:digital teaching tool, kindergarten, lower grades, number concept

Introductory

Taking into account the latest research results, we need to look at the role of vision, hearing and touch in learning differently (Kovács 2010, Csépe 2007). Although each of them has many possibilities individually, most of the "proven" competence development methods involve the joint processing of information from multiple senses. Since the effectiveness of one sense organ may even weaken the effectiveness of another information acquisition channel, it is important to choose the appropriate proportions. The decision is determined not only by the cognitive abilities of the students, but also by the content of the curriculum. During mathematics teaching – according to Bruner's representation theory – knowledge is communicated in three (material, iconic

[22]not speech, but auditory chains with a logical structure (the basic unit of auditory perception) Csépe (2007)

or symbolic) plane. We cyclically return to the same mathematical concept at different levels of development.
We aim to support the understanding of the fundamentals of the concept of function (number concept, sequence formation, assignment, and orientation on the plane, which is essential for interpreting graphs) with the help of the "Audible Mathematics" software package.
This introduction presents the teaching tool, which is still being tested, for the first time, so the focus is not on equal opportunities, but on attracting the attention of large-group kindergarten teachers and teachers working with sighted children.

1. Methodological and ergonomic aspects taken into account during development

Our goal is to expand the multi-sensory school environment that supports the development of basic mathematical competencies in 4-7 year old children (even those blind from birth) with ICT tools in order to reduce inequalities of opportunity.
In simple terms, the abstract nature of mathematics means that objects (objects, events) are less important in mathematics, and the relationships between them are much more important – the logical structure. At the level of specific operations, the description of the structure is made up of, for example, logical characteristics that can be demonstrated in grouping, classification or sequencing. According to Piaget's theory, the majority of 4-7 year olds are still in the pre-operational stage of development. Their thinking is not wrong, it's just different. We see the task of developing abilities not in practicing as much as possible with the child, but in providing a space appropriate to their individual level for collecting experience. We need a learning environment that integrates the possibility of developing thinking in action, and playful activities during which the child can experience, for example, that the set and its number do not depend on the location or order of its elements. In our opinion,*action modeling*is the most complicated process in the development of traditional teaching aids. For example, on a printed worksheet, the addition (adding to 10) has to be solved by drawing or coloring objects, in our software we move the objects using a “drag and drop” technique using a mouse.

With the help of the "Audible Mathematics" software package:

- we can move virtual objects on the computer screen with a mouse (or even by touching the surface of the digital tablet),
- in write mode we can write out the relational symbols or the mathematical expression of the event,
- we can connect elements,
- we can predelimit the part of the board where a predetermined number of elements must be counted, etc.

You can observe in the dynamics of the child which object manipulations correspond to the set operations. If we work with a digital whiteboard, we can not only divide the whiteboard into separate areas to compare different cases, but we can also save them[23]The following novel solution*age-specific body image*[24] was encouraged by the consideration of the number concept. More and more experts dealing with the development of the number concept see the reason for the underdevelopment of counting competence in the inadequate perception of parts and wholes (Csocsán 2010). According to the literature, although a 4-5 year old child recognizes certain details, he grasps them in isolation, without their connection with the whole. Overcoming this is especially difficult later in the case of visually impaired students (Pálhegyi 1980). Hearing can help us. The perception of the whole together with its details has even been given a special name in psychoacoustics – the Mozart effect. Since the ability of objects to make sounds is also a property for children, the next solution, which has gained a right to exist among students blind from birth, was obvious. If we assign a sound to each object (e.g., the Do sound obtained by pressing a piano key to an apple, and the Mi sound to a pear), then the rhythmic Do-Do, Mi-Mi, Do-Mi or Mi-Do, interspersed with short pauses, heard in pairs, not only reproduces the "image" of the fruits separated by their pairs, but also highlights the properties of the structure that are essential for understanding even-odd, and multiplication as a multiple addition operation. This is no coincidence. The brain tries to harmonize the seen and heard images in a unique way – it instinctively "predicts" the continuation of the expected sound. Many researchers share the opinion that the Gestalt principles that are implemented during the interpretation of the seen image also apply when processing the heard information.

[23]Unfortunately, we can only capture the static image when saving.

[24]By form perception we mean the perception of related wholes as a unified whole

In an effort to synchronize the image and audio, the software allows you to access audio samples by clicking on objects. This not only "accentuates" the image, but also the object manipulation and mathematical structure (e.g. Dó-Re-Mi – counting to three, Dó-Dó—Dó-Dó-Mi — Dó-Dó—Dó-Dó-Mi — Dó-Dó—Dó-Dó-Mi – grouping by five).

The software also contains live speech. Since live sound can be accompanied by metacommunication elements, it can have a greater impact. The time factor is dominant in the processing of sound information, and depending on their form of appearance, we can also influence emotions and the subconscious. By playing sound samples, we are able to more easily influence the attention of students (reduce noise in the visual space and highlight the part from the whole).

The usability of the software is not limited to frontal information delivery for the purpose of illustration. Our primary goal is to motivate students to be active. If a child is hindered in manipulating objects, they may still be able to work with a touch screen. In the case of students with severe visual impairment, the sound samples played can even guide their activities with objects. Using a keyboard, they can make their own solution visible to others, thereby creating a working relationship with others to a certain extent.

The software package does not include an evaluation system.

2. Brief description of the software

1. Runtime environment

THE free used teaching aids are available the http://fano.tsf.hu/hallhato-matematika/ website. The separate software components are suitable for both assisting teachers in their work during school education and for use at home in a playful setting with parental cooperation.

The elements of the software package were developed with GameMaker8 game creation software and are operating system independent (*.exe executable files). Their size ranges from 3-15 MB. Recommended screen resolution: at least 1024x768. When using them, the computer can be controlled with both the left and right mouse buttons (the right mouse button has a separate function) and the keyboard. In the case of a digital whiteboard, it is important to check the availability of the functions programmed for the right mouse button. For students with severe visual impairments, it is recommended to install the Jaws for Windows text-to-speech software. The unused

It is advisable to remove the keys in order to improve spatial orientation (Báthori 2008).

2. User Guide

The playful software – regardless of methodology – is recommended for the kindergarten age group and the first semester of the first grade (max. in the number circle of 8). In addition to the software package, additional tools recommended for sighted students: red-blue counting disc (box), colored plastic sticks, colored rod set. Depending on the use, each software component is suitable for the development of several basic competencies. We aim to support orientation in the plane with the games under the "Table" menu item in the szamegyenes_elokeszitese2.exe and darabszam_sor_oszlop.exe software.

More detailed software descriptions and usage suggestions for teachers can be found on the above-mentioned website. We would like to arouse the interest of our kindergarten teachers by describing the following few typical tasks.

Basic mathematical competence: recognition–identification of direct and previous experience.

Type tasks:

- with a pair, play the elementary sound patterns directly one after the other, and if the second member of the pair is not the same as the first, the child must indicate
- Click on the specified fruit or number.

*Software component:*each.

Basic mathematical competence: selection–separating a part of an existing set based on a given or chosen criterion. *Type tasks:*

- Find the keys on the keyboard that make the same sound.
- let's just check the apples
- Collect the apples placed by the adult using the "grab and drag" technique to a place designated by them.

Software component: B_selection_5.exe, hallasdev.exe, selection_apple_peach_selection.exe, set operations.exe. Basic mathematical competence:*line creation* –arranging objects in order according to a specific rule.

Type tasks: children first work based on verbal instructions and the sound samples are only present as accompanying elements, later only

The objects are arranged from left to right or bottom to top according to the rule defined by the sound sample sequence.
Software component: serial_builder_apple_barack_korte.exe, soralkotas_sikidomok.exe.
Basic mathematical competence: a*rulemaking*–verbal formulation of recognition.
Type tasks:

-verbally stating the rules of the pattern line started by the adult and continuing the line creation

-Creating a line that is different from the sample line created by the adult after communicating the difference verbally.

Software component: serial_samples.exe, soralkotas_alma_barack_korte.exe, soralkotas_sikidomok.exe.

Abstrack
The Presentation of the Software Package "Audible Mathematics" By completing visually impaired students' tools with audible information we can cause a positive effect not only on the development of hearing and attention as well as the ability to analyze heard information, but on the development of numerous basic mathematical skills as well.

With the help of the softwares we have developed, we would like to expand the number of tools which support the integrated education of blind and visually impaired pupils. While we were developing the digital tools which can also be used by the normally sightedlowerprimary school pupils, we fully tried to take into account the experience gained in the primary target group, the most recent results of psychoacoustics and the current didactic methodologies.

Keywords:digital tools, kindergarten, primary school, number concept

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